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ESP32 SPEECH FUNCTION: Text To Speech

SOMNATH BERA

Speech capability in technology primarily has two dimensions: text to speech (TTS) and speech to text (STT). This device focuses on TTS. Here, the MAX98357A amplifier is used with the MCU ESP32 development board, and Google API is used to convert the text into speech.

BILL OF MATERIALS	
Components	Quantity
ESP32 node MCU (MOD1)	1
3-watt speaker	1
MAX98357A I2S 3-watt Class D amplifier	1
5V USB power supply cable	1

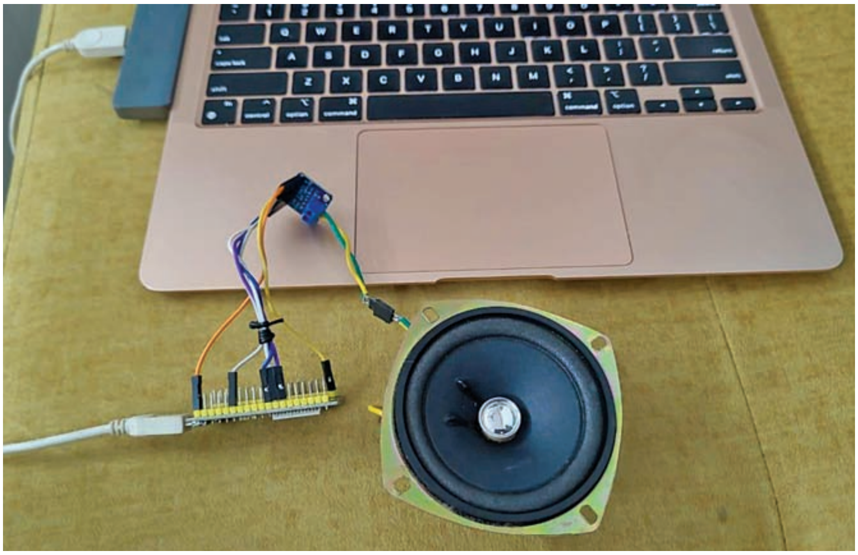

Fig. 1: Author's prototype

The cost-effective MAX98357A I2S amplifier (mono) is connected to the ESP32. Although stereo models like the UDA1334A are available, the mono version is chosen for simplicity. The I2S amplifier uses three GPIO pins, which can be any pins except for 34 and 35 (input-only pins). A 4-ohm speaker is connected to the output, and it is essential to verify the board specifications. The speaker's ' + ' and '-' terminals must be connected correctly to avoid distortion. The author's prototype is shown in Fig. 1, and the necessary components are listed in the Bill of Materials table.

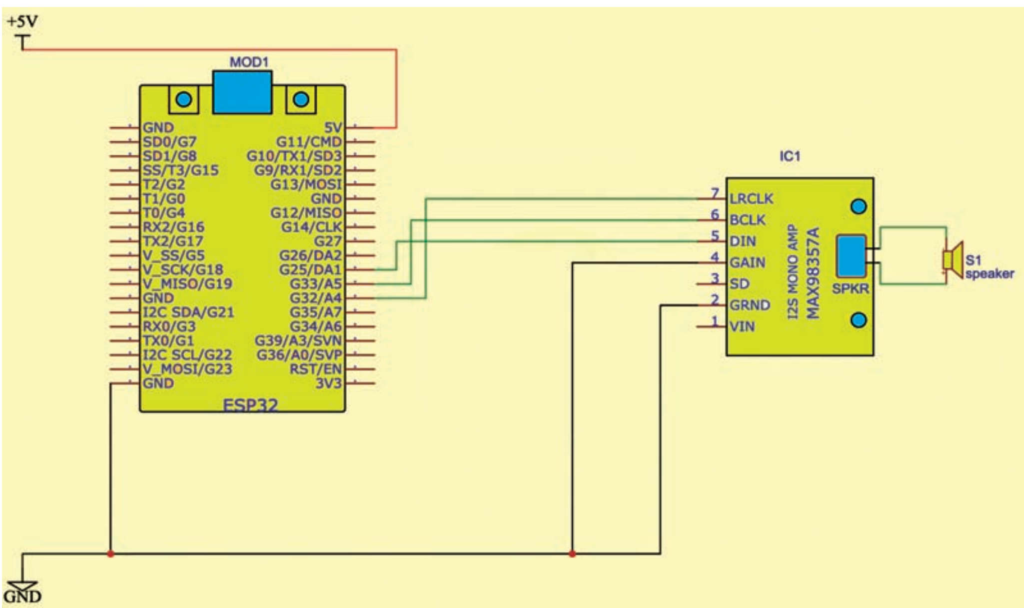

Fig. 2: Circuit diagram of ESP32 function

Circuit diagram

The circuit diagram of the ESP32 speech function is shown in Fig. 2. It is built around the ESP32 Node MCU (MOD1), MAX98357A